



BridgeSim: Unveiling the OL-CL Gap in End-to-End Autonomous Driving

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Abstract: Open-loop (OL) to closed-loop (CL) gap (**OL-CL gap**) exists when OL-pretrained policies scoring high in OL evaluations fail to transfer effectively in closed-loop (CL) deployment. In this paper, we unveil the root causes of this systemic failure and propose a practical remedy. Specifically, we demonstrate that OL policies suffer from *Observational Domain Shift* and *Objective Mismatch*. We show that while the former is largely recoverable with adaptation techniques, the latter creates a structural inability to model complex reactive behaviors, which forms the primary OL-CL gap. We find that a wide range of OL policies learn a biased Q-value estimator that neglects both the reactive nature of CL simulations and the temporal awareness needed to reduce compounding errors. To this end, we propose a Test-Time Adaptation (TTA) framework that calibrates observational shift, reduces state-action biases, and enforces temporal consistency. Extensive experiments show that TTA effectively mitigates planning biases and yields superior scaling dynamics than its baseline counterparts. Furthermore, our analysis highlights the existence of blind spots in standard OL evaluation protocols that fail to capture the realities of closed-loop deployment.

Keywords: Autonomous Driving, Closed-loop Simulation, Test-time Adaptation

1 Introduction

A growing number of works in end-to-end (E2E) autonomous driving have directed their efforts toward achieving high scores on open-loop (OL) benchmarks such as nuScenes [1] and NAVSIM [2, 3], with some reporting performance at or near human level [4, 5]. However, the trajectories output from OL policies frequently become infeasible or unsafe when deploying in closed-loop (CL) simulation, which shows strong discrepancy with their claimed behaviors under OL evaluations [6, 7, 8, 9]. We refer to such a discrepancy as the **OL-CL gap** in E2E driving.

Unveiling the OL-CL gap matters for avoiding hallucinated conclusions from OL benchmarks. We illustrate our motivations in Fig. 1. First, success in OL benchmarks might create a misleading proxy for driving behavior in realistic conditions. Recent works [6, 7, 8] have demonstrated a lack of correspondence between OL and CL performance under similar domain settings. Furthermore, this correlation decays as the simulation horizon increases (see Fig. 3). Second, OL scaling laws fail to predict CL scaling. While recent studies show that OL performance scales predictably with data volume and test-time compute, neither of these relationships transfers reliably to CL deployment [10, 11]. These discrepancies suggest that OL evaluation protocols, while being a practical and scalable proxy for costly CL simulation [1, 12, 3, 2, 13], might be biased and overlook the underlying objectives that real-world driving policies should optimize.

In this paper, our goal is not to criticize OL evaluation protocols or the development of methods drawn from OL evaluations, but rather to clarify *where* policies might fail and *what* adaptations are needed to mitigate failures in CL deployment. Our key contributions are summarized as follows:

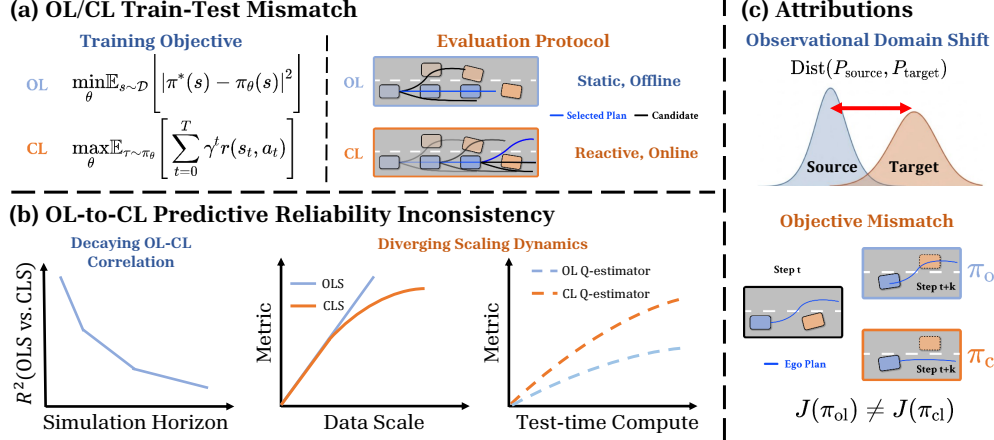


Figure 1: **Motivation.** (a): Open-loop (OL) train-test paradigms do not necessarily align with the objectives and evaluation metrics in closed-loop (CL) deployment; (b): This mismatch results in significant performance discrepancies, characterized by decaying correlation and diverging scaling dynamics; (c): We attribute this OL-CL gap to *observational domain shift* and *objective mismatch*.

- 38 **1. Identify the fundamental failure modes of the OL-CL gap in E2E autonomous driving.** We
39 attribute the OL-CL gap to two sources. The first is *Observational Domain Shift*, where test-time
40 sensor observations differ from training observations, degrading perception before planning. While
41 incomplete observability might bring planning failure, we show that this cross-domain gap can be
42 mitigated through domain adaptation, which recovers the required observability for planning. How-
43 ever, perception calibration alone does not prevent planning failure in closed-loop simulation. The
44 second, more fundamental source is *Objective Mismatch*: OL training optimizes static trajectory-
45 matching objectives that do not correspond to true CL returns, yielding biased Q-value estimates in
46 CL deployment. Such bias is structural and persists even when perceptual inputs are well calibrated.
- 47 **2. Propose a test-time adaptation (TTA) framework that mitigates CL deployment failure.**
48 The proposed TTA framework has two components: a *flow-matching observational calibrator* for
49 cross-domain latent feature alignment, and a *training-free policy adaptation mechanism* for miti-
50 gating structural objective mismatch. The latter combines a *truncated Q-value estimator* to reduce
51 OL-induced scoring bias with an *adaptive replan mechanism* for reactive execution under evolving
52 dynamics. Together, these modules mitigate OL-to-CL deployment failures. Across diverse CL
53 scenarios, our method improves baseline performance by up to 22% and exhibits stronger test-time
54 scaling than unadapted policies.
- 55 **3. Introduce BridgeSim, a cross-simulator E2E closed-loop simulation platform.** To com-
56 plement existing E2E benchmarks, we develop BridgeSim designed for the rigorous, systematic
57 diagnosis and evaluations of any E2E driving policy under interactive, closed-loop settings.

58 2 Related Work

59 **Driving Simulators and Benchmarks.** Driving simulators provide safe, repeatable environ-
60 nments for training and evaluating autonomous driving systems [14, 15, 16, 17, 18]. Tradition-
61 al game-engine and high-fidelity platforms, including GTA V [19], Sim4CV [20], AIR-
62 SIM [21], and CARLA [22], support configurable scenes and complex physical conditions, while
63 lightweight or modular platforms such as DeepDrive [23], DriverGym [24], and MetaDrive [25]
64 emphasize efficient experimentation, scalability, and multi-agent interaction. However, these
65 simulators often exhibit domain gaps from handcrafted assets and limited visual realism. Re-
66 cent data-driven and generative simulators address this issue through realistic scene synthesis
67 and reconstruction: DriveArena [26] enables iterative perception-action loops via world models;
68 HUGSIM [27] and RAD [28] use 3D Gaussian Splatting for photorealistic closed-loop simulation;
69 and ReconDreamer/Recondreamer-RL [29, 30] use world-model-based reconstruction for novel tra-

jectories. Nevertheless, existing generative simulators still suffer from artifacts and lack metric annotations for evaluating E2E policies (see Appendix E).

End-to-end Autonomous Driving. Traditional autonomous-driving stacks decompose perception, prediction, and planning, but suffer from error propagation and information loss [31, 32, 33, 34]. This has motivated end-to-end (E2E) policies that map sensor inputs directly to driving actions [35, 36]. Recent methods are built from TransFuser [37] backbone and further improve action modeling: DiffusionDrive/DiffusionDriveV2 [38, 39] capture multi-modal action distributions with diffusion policies, while DrivoR [5] uses register tokens to avoid complex intermediate representations. Despite showing strong OL metrics, AD-MLP, PDM-Close, BEV-Planner, LEAD, and RAP [40, 41, 7, 42, 4] reveal the evaluation gaps of OL benchmarks indicating CL performance and the need to bridge the OL-CL gap. Although CL training can mitigate this gap, it incurs substantial complexity and computation [8]. In this paper, we analyze such OL-CL gap and address them via the lens of test-time adaptation.

Table 1: Comparisons with other E2E open-loop and closed-loop benchmarks. MV-Consistency: multi-view temporal consistency, TF-Consistency: Traffic elements annotations consistency. PG: procedural map and behavior generation. See Appendix B for more details of BridgeSim platform.

Name	Closed-loop	Long-term Simulation	MV-Consistency	TF-Consistency	Supported Maps	Supported Traffic Modes
NAVSIM [3, 2]	×	×	✓	✓	nuPlan	Log-replay
Bench2Drive [6]	✓	✓	✓	✓	CARLA	IDM
DriveArena [26]	✓	✓	×	×	nuScenes	Log-replay
HUGSIM [27]	✓	✓	✓	×	nuScenes, Waymo, KITTI-360, PandaSet	Log-replay, IDM, Adversarial
BridgeSim (ours)	✓	✓	✓	✓	CARLA, Waymo, nuScenes, nuPlan, PG	Log-replay, IDM, Adversarial

3 BridgeSim: A Unified Cross-Simulator Closed-loop Simulation Platform for E2E Driving Policies

To empirically investigate the OL-CL gap with quantifiable rigor, we introduce BridgeSim, a cross-simulator platform designed to evaluate OL pretrained policies within high-fidelity CL environments. BridgeSim designs a unified scenario protocol to incorporate diverse map scenarios (e.g., nuPlan [12], WOMD [43], and nuScenes [1]) with heterogeneous traffic modes (e.g., log-replay, IDM [44], and adversarial policies [45]) to stress-test E2E driving policies under a closed-loop simulation environment. Furthermore, BridgeSim offers a flexible deployment setting to simulate open-loop policy with varying execution frequencies and simulation horizons. Consequently, BridgeSim bridges the critical divide between OL benchmarks limited to short-term static prediction and existing CL benchmarks that often lack the comprehensive functionalities and annotations required for complex, reactive stress-testing. A comprehensive functional comparison between BridgeSim and other prominent E2E benchmarks is provided in Table 1.

4 Decomposing the Open-Loop to Closed-Loop (OL-CL) Gap

In this section, we revisit the theoretical formulation of the E2E driving task and analyze the failure modes of OL-CL gap, namely *observational domain shift* and *objective mismatch*. Through empirical studies, we further illustrate the performance drop caused by those factors, which motivate the design of a test-time adaptation framework that recovers the OL policies from failures.

4.1 Preliminaries and Notations

POMDP Formalism. We formalize the end-to-end driving task as a Partially Observable Markov Decision Process (POMDP), defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{T}, \Omega, r, \gamma, \rho_0, T \rangle$. Here, $\mathcal{S}$ denotes the true underlying state space of the environment, $\mathcal{A}$ is the ego-vehicle’s action space, and $\mathcal{O}$ is the high-dimensional observation space. At time step t , the system occupies state $s_t \in \mathcal{S}$. The agent receives an observation $o_t \in \mathcal{O}$ governed by the emission distribution $\Omega(o_t | s_t)$, executes an action $a_t \in \mathcal{A}$, and the environment evolves via $\mathcal{T}(s_{t+1} | s_t, a_t)$. The episode proceeds for a finite horizon of T steps, where $\rho_0(s_0)$ represents the initial state distribution and $\gamma \in [0, 1)$ is the discount factor.

End-to-End Driving Policy. We model an end-to-end (E2E) driving policy as a stochastic mapping $\pi_{\theta, \phi}^d$ under observation domain $d \in \{\text{source}, \text{target}\}$, with encoder parameters θ and behavior decoder parameters ϕ . At time t , the policy predicts an action sequence¹ $\mathbf{a}_t \in \mathcal{A}$ from an observation $o_t \sim \Omega^d(\cdot | s_t)$ of the latent state s_t . This domain-dependent emission model isolates observational domain shift from behavioral shift. The encoder $f_\theta : \mathcal{O} \rightarrow \mathcal{Z}$ maps o_t to a latent scene representation z_t , and the policy head $\pi_\phi(\mathbf{a}_t | z_t)$ outputs the ego future trajectory. The induced joint distribution factorizes as

$$\pi_{\theta, \phi}^d(\mathbf{a}_t, z_t, o_t | s_t) \triangleq \pi_\phi(\mathbf{a}_t | z_t) P_\theta(z_t | o_t) \Omega^d(o_t | s_t), \quad (1)$$

where $P_\theta(z_t | o_t)$ is the encoder-induced conditional distribution. For deterministic encoders commonly used in recent E2E planners [38, 39, 4], $z_t = f_\theta(o_t)$ and thus $P_\theta(z_t | o_t) = \delta(z_t - f_\theta(o_t))$.

Closed-loop Objective. We seek to maximize the performance of a driving policy π via the expected closed-loop cumulative return $J(\pi) \triangleq \mathbb{E}_{s_0 \sim \rho_0, \tau \sim \pi} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t) \right]$, where the reward $r(s_t, a_t)$ evaluates the *executed* control a_t , which typically constitutes only a partial subset of the full sequence $a_{t:t+H}$ generated by the policy at each time step.

4.2 Observational Domain Shift

Failure mode analysis. Observational domain shift occurs when observation encoder fails to extract coherent semantic features (e.g., lane, objects, etc.) in the target environment and impairs the *state observability* required for planning, as illustrated in Fig. 2(a). To denote the impact of this domain shift on task performance, we define the *Observational Domain Gap* Δ_{obs} as:

$$\Delta_{\text{obs}}(\theta, \phi) \triangleq J(\pi_{\theta, \phi}^{\text{source}}) - J(\pi_{\theta, \phi}^{\text{target}}), \quad (2)$$

where an identical control head π_ϕ isolates the degradation in expected return attributable solely to domain shift. The discrepancy between source and target domains can be measured by:

$$\text{Dist}_{\mathcal{Z}}(s; \theta) \triangleq \text{Dist}\left(P_\theta(z | o) \Omega^{\text{source}}(o | s), P_\theta(z | o) \Omega^{\text{target}}(o | s)\right), \quad (3)$$

where $\text{Dist}(\cdot, \cdot)$ is the distance metric over distributions on $\mathcal{Z}$. With this formulation, we hypothesize that observational domain gap is recoverable as long as we calibrate the observation encoder θ to minimize the distributional distance.

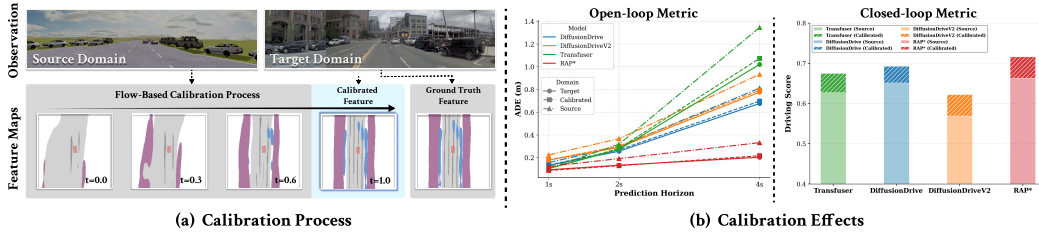


Figure 2: **Empirical Study on Observational Domain Shift.** (a): Flow-matching procedure for observational calibration; (b): Calibration effects on OL and CL metrics. See Appendix D.2 for experimental details.

Empirical studies. To test this hypothesis, we isolate Δ_{obs} through an open-loop ablation that decouples observational errors from closed-loop compounding effects such as control drift; see Fig. 2(b). Following [46], we vary only the input observation domain while keeping state transitions fixed, and measure trajectory deviation from the expert. We find that observational shift substantially degrades policy performance because the perception module fails to recover planning-relevant semantics. However, our calibration method in Section 5.1 restores these scene features and reduces the gap. These results suggest that **observational domain shift primarily degrades perception before planning, making it largely recoverable through targeted calibration of state observability.**

¹We use bold $\mathbf{a}_t$ to denote the control sequence $a_{t:t+H}$, where H is the fixed prediction horizon set during training.

4.3 Objective Mismatch

Failure mode analysis. Objective mismatch arises because OL training optimizes a *static full-sequence objective*, whereas CL deployment follows a *reactive receding-horizon process*: the agent predicts an H -step trajectory but executes only a k -step prefix before receiving a new observation and replanning. We formalize this with a unified closed-loop objective parameterized by execution horizon k :²

$$J_k(\pi) = \mathbb{E}_{s_0 \sim \rho_0, \tau \sim \pi_k} \left[\sum_{c=0}^{T/k-1} \gamma^{ck} \sum_{i=0}^{k-1} \gamma^i r(s_{ck+i}, a_{ck+i}) \right], \quad (4)$$

where π_k replans every k steps and c indexes planning cycles. This formulation decomposes OL-to-CL degradation from objective mismatch into two factors: **biased Q-value estimation** and **compounding execution error**.

Biased Q-value estimation stems from OL training, where E2E planners minimize errors over the full predicted horizon H . This effectively optimizes $J_{OL}(\pi) \approx J_H(\pi)$, implicitly setting the execution horizon k equal to the planning horizon H . However, $J_H(\pi)$ unrolls future states under open-loop kinematic assumptions, ignoring environment stochasticity, covariate shift, and reactive traffic behaviors. As a result, the induced offline proxy distribution is misaligned with the true closed-loop state-visitation distribution $d^{\pi_{\theta}, \phi}$ generated by high-frequency active execution.

Beyond biased Q estimation, OL training also lacks the temporal robustness required for sequential decision-making, making policies vulnerable to compounding execution errors. Since the policy is not optimized for stability under consecutive high-frequency executions, the vanilla k -step execution strategy often fails to maximize CL return, as shown in Fig. 3(a). We define the resulting *Objective Mismatch Gap* as

$$\Delta_{\text{obj}}(\theta) \triangleq J_{k^*}(\pi_{\theta, \phi_{\text{CL}}}) - J_{k=H}(\pi_{\theta, \phi_{\text{OL}}}), \quad (5)$$

where ϕ_{CL} is optimized under the unified CL objective with optimal execution horizon k^* over the induced distribution $d^{\pi_{\theta}, \phi}$, while ϕ_{OL} is optimized only under the OL proxy objective ($k = H$) on the offline expert distribution. This formalization suggests that objective mismatch is a structural barrier to CL transfer: the OL proxy induces optimization targets that need not maximize CL return.

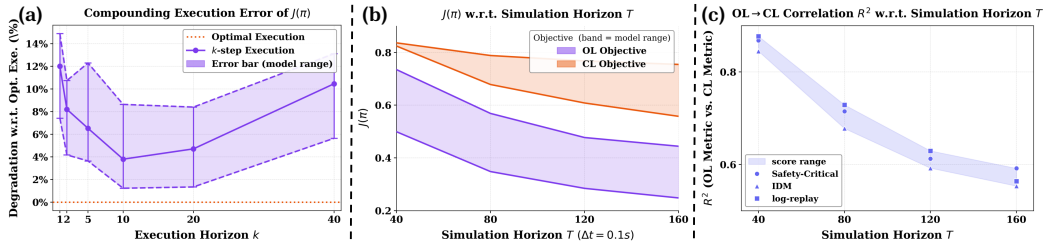


Figure 3: **Objective Mismatch.** (a) Compounding execution error is sensitive to execution horizons k . (b) CL-OL objective divergence, defining Δ_{obj} . (c) OL-CL Pearson correlation across traffic modes and simulation horizons T . Details are in Appendix D.3.

Empirical studies. We characterize the objective mismatch Δ_{obj} over simulation horizon T across representative E2E policies [38, 39, 37, 4]; see Fig. 3(b). The divergence between OL and CL returns grows sharply once $T > H$, indicating that OL Q-value estimates become unreliable for long-horizon interaction. Fig. 3(c) further shows that the Pearson correlation between OL and CL metrics decays substantially when $T > H$, suggesting that OL evaluation can misrepresent CL deployment readiness. These results identify objective mismatch as a primary source of the OL-CL gap and motivate test-time adaptation to correct biased Q estimates and improve sequential decision-making during CL execution.

²We restate this formulation to highlight the role of execution horizon k . The standard closed-loop objective in Section 4.1 is recovered as the special case $k = 1$.

5 Proposed Method: Test-time Adaptation (TTA) Framework

Motivated by the decomposition analysis in Section 4, we propose a test-time adaptation (TTA) framework that improves closed-loop robustness of OL-pretrained E2E policies without additional end-to-end retraining. The framework consists of an observational calibrator that recovers latent representations via flow-matching (Section 5.1) and a policy adaptation procedure to mitigate biased Q-value estimation and adaptively perform execution to maximize final CL return (Section 5.2).

5.1 Observational Calibrator

Following Section 4.2, we define observational shift through the latent distributions induced by a fixed pretrained encoder $f_\theta : \mathcal{O} \rightarrow \mathcal{Z}$. For each domain $d \in \{\text{source}, \text{target}\}$ with state distribution ρ^d and observation channel $\Omega^d(\cdot | s)$, let $\mathcal{D}_d$ denote the marginal law of $z = f_\theta(o)$ under $s \sim \rho^d$ and $o \sim \Omega^d(\cdot | s)$. Although $\mathcal{D}_{\text{source}}$ and $\mathcal{D}_{\text{target}}$ share support in $\mathcal{Z}$, they differ due to domain-dependent sensing.

During training, we learn an observational calibrator $g : \mathcal{Z} \rightarrow \mathcal{Z}$ via flow matching such that $g_\# \mathcal{D}_{\text{source}} \approx \mathcal{D}_{\text{target}}$ [47], mapping source-domain features into target-compatible representations for the downstream policy π_ϕ . Rather than using a vanilla flow-matching objective, we adopt a multi-objective calibration loss inspired by [48]. Fig. 2 shows that this calibration aligns latent features and recovers both OL and CL metrics. See Appendix C.1 for details.

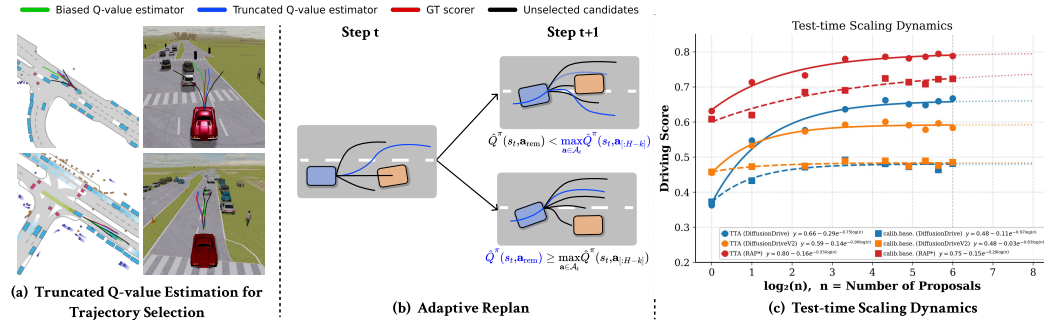


Figure 4: **Test-time Policy Adaptation.** (a): Truncated Q-value estimator selects safe planned trajectories during closed-loop execution; (b): Adaptive replan preserves previously verified planned trajectories if the newly proposed trajectory doesn't obtain higher Q-values; (c): Test-time scaling dynamics between biased (baseline) and truncated Q-value estimators (TTA).

5.2 Test-time Policy Adaptation

5.2.1 Truncated Q-value Estimation

To mitigate biased Q-value estimates learned from open-loop proxy objectives, we introduce a dynamic test-time truncated action-value estimator. At decision step t , the stochastic policy $\pi_{\theta, \phi}^\Omega(\cdot | s_t)$ samples N candidate trajectory-actions $\mathcal{A} = \{\mathbf{a}^{(1)}, \dots, \mathbf{a}^{(N)}\}$, where each $\mathbf{a}^{(i)}$ is a spatio-temporal plan over horizon H .

Standard prefix evaluation combines the k -step execution reward with the future value $\gamma^k \mathbb{E}[V^\pi(s_{t+k})]$, but this implicitly depends on rewards beyond the explicit planning horizon. Estimating such long-horizon returns with an open-loop transition model is unreliable due to compounding covariate shift and unmodeled environment stochasticity. We therefore restrict evaluation to the reliable planning window by defining

$$\hat{Q}^\pi(s_t, \mathbf{a}_t) \triangleq R_k(s_t, \mathbf{a}_t) + \gamma^k \mathbb{E}[V^\pi(s_{t+k})] - \gamma^H \mathbb{E}[V^\pi(s_{t+H})], \quad (6)$$

where $R_k(s_t, \mathbf{a}_t) = \sum_{i=0}^{k-1} \gamma^i r(s_{t+i}, \mathbf{a}_{t+i})$ captures geometric and historical penalties over the executed prefix, and $\gamma \in [0, 1)$ is the discount factor. The subtraction term cancels the intractable tail beyond H , yielding a horizon-bounded estimator aligned with the model's explicit plan. Appendix C.2 details the expected Q-function implementation.

As shown in Fig. 4(a), closed-loop execution selects the candidate maximizing $\hat{Q}^\pi$ rather than the biased open-loop score [39, 38]. Repeating this selection at each step dynamically filters trajectories affected by the objective mismatch in Section 4.3.

5.2.2 Adaptive Replan

Standard test-time scorers select only from the current candidate set $\mathcal{A}_t$, leading to high-frequency action chattering and temporal inconsistency across planning cycles. To reduce compounding execution errors, we preserve the unexecuted portion of the previously verified plan and replan only when necessary, as shown in Fig. 4(b).

Let $\mathbf{a}_{\text{prev}}^*$ be the trajectory selected at step $t - k$. After executing its first k steps, we transform the remaining $H - k$ waypoints into the current ego-centric frame at s_t and denote the resulting persistent plan as $\mathbf{a}_{\text{rem}}$. For fair comparison with new candidates $\mathbf{a} \in \mathcal{A}_t$, we evaluate $\mathbf{a}_{\text{rem}}$ against the corresponding $(H - k)$ -step prefixes $\mathbf{a}_{[:H-k]}$ using the truncated estimator in Eq. 6.

The selected plan is

$$\mathbf{a}_t^* = \begin{cases} \mathbf{a}_{\text{rem}} & \text{if } \hat{Q}^\pi(s_t, \mathbf{a}_{\text{rem}}) \geq \max_{\mathbf{a} \in \mathcal{A}_t} \hat{Q}^\pi(s_t, \mathbf{a}_{[:H-k]}), \\ \arg \max_{\mathbf{a} \in \mathcal{A}_t} \hat{Q}^\pi(s_t, \mathbf{a}) & \text{otherwise.} \end{cases} \quad (7)$$

Thus, the agent keeps the persistent plan unless a newly sampled candidate achieves a strictly higher estimated return over the shared horizon.

6 Experiments

6.1 Experimental Settings

We leverage BridgeSim as the closed-loop simulator for experiments (see Appendix B for platform details). We evaluate TTA framework on pretrained E2E planners from NAVSIM [2], including DiffusionDrive [38], DiffusionDriveV2 [39], and RAP [4]. Experiments cover three settings: 1) **in-domain evaluation**, where models are tested on NavHard log-replay scenarios under varying simulation horizons; 2) **unseen scene evaluation**, where models are tested in log-replay scenarios from domains outside of pretraining, such as nuScenes [1] and WOMB [43]; and 3) **reactive evaluation**, where log-replay agents are replaced by IDM [44] or adversarial agents [45]. Closed-loop performance is measured by the *BridgeSim Driving Score (DS)*, which combines Extended Predictive Driver Model Score (**EPDMS**) and Route Completion (**RC**). **EPDMS** further aggregates: No At-Fault Collisions (**NC**), Drivable Area Compliance (**DAC**), Traffic Light Compliance (**TLC**), Driving Direction Compliance (**DDC**), Lane Keeping (**LK**), Time-to-Collision (**TTC**), History Comfort (**HC**), and Extended Comfort (**EC**). Appendix B.4 details the metrics, and Appendix D.1 provides additional experimental details.

6.2 Main Results

In-domain evaluation. Fig. 5(a) reports log-replay results on BridgeSim-NavHard across simulation horizons of 4/8/12/16 seconds. E2E planners generally degrade as the horizon increases due to compounding covariate shift. In contrast, TTA consistently flattens this decay across all baselines by mitigating *Objective Mismatch* during closed-loop execution. These results show that TTA improves temporal consistency and safety in long-horizon CL simulation.

Unseen-scene evaluation. Fig. 5(b) shows that TTA consistently improves diverse OL baselines [38, 39, 4] across NavHard [2], nuScenes [1], and WOMB [43], demonstrating cross-domain robustness.

Reactive evaluation. Table 2 shows that TTA improves key CL metrics across traffic modes and baseline architectures, including progress-related **RC** and safety-related **TTC/NC**. These gains indicate that TTA improves efficiency while preserving safety in reactive simulations.

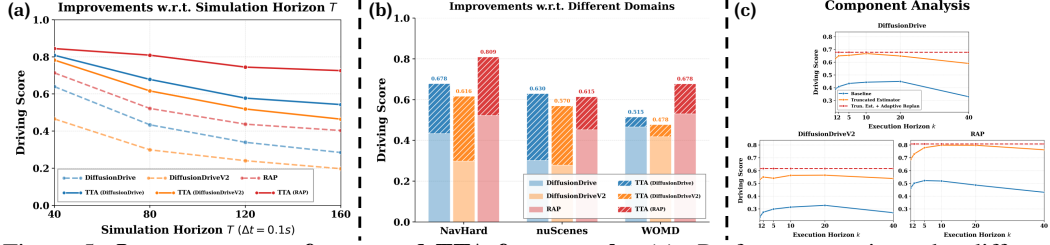


Figure 5: **Improvements of proposed TTA framework.** (a): Performance gain under different simulation horizons on BridgeSim-NavHard scenarios; (b): Performance gain under different log-replay domains; (c): Component analysis of TTA framework.

246 **Component analysis.** Fig. 5(c) isolates each TTA module. The truncated Q-value estimator con-
 247 sistently outperforms the biased baseline across execution horizons k , while adaptive replan further
 248 mitigates compounding execution errors, yielding more stable driving scores across backbones.

249 **Test-time scaling dynamics.** Following [38, 39, 49, 50], we study test-time scaling by increasing the
 250 number of trajectory proposals. As shown in Fig. 4(c), scaling behavior in CL simulation strongly
 251 depends on scorer quality. Our proposed scorers consistently benefit from larger proposal sets and
 252 broader state visitation, whereas biased Q-value estimators show limited gains.

Table 2: Performance comparisons with different reactive traffic modes. Note that TTA improves key closed-loop metrics such as NC, TTC, and RC, resulting in overall higher DS. See Appendix D.4 for complete sub-metric scores.

Model	IDM							Safety-Critical						
	DS	EPDMS	RC	NC	DAC	TTC	LK	DS	EPDMS	RC	NC	DAC	TTC	LK
DiffusionDrive	45.30	72.95	59.93	95.41	85.47	89.73	89.83	38.68	68.16	54.77	93.62	83.86	84.97	90.71
DiffusionDriveV2	31.94	65.43	45.85	94.57	82.09	89.10	85.42	28.17	61.45	43.50	93.15	80.78	85.72	85.81
RAP	60.02	76.85	77.20	97.68	88.84	93.07	78.16	51.83	70.88	71.22	95.03	88.16	86.37	78.26
TTA (DiffusionDrive)	62.68 +17.4	84.36	72.28	96.85	95.63	91.57	92.43	55.02 +16.3	78.95	67.92	94.73	95.21	86.64	91.30
TTA (DiffusionDriveV2)	53.96 +22.0	77.47	67.89	96.08	90.53	89.46	89.98	47.64 +19.5	71.66	65.91	93.54	89.16	84.23	90.11
TTA (RAP)	79.07 +19.1	90.92	86.21	98.28	98.62	95.31	91.52	65.94 +14.1	82.01	78.66	95.30	96.19	87.71	90.26

253 7 Limitations

254 **Sim2Real Gap Interpretation.** Our experiments do not include on-road deployment, since real-
 255 world evaluation at the scale remains prohibitively expensive and unsafe. We do not argue that
 256 absolute closed-loop scores transfer directly to real driving, but position BridgeSim as a controlled
 257 hypothesis-testing platform whose structural conclusions describe properties of E2E driving systems
 258 themselves rather than artifacts of a particular simulator.

259 **Scenario Evaluation Coverage.** Our experiments are conducted on dense urban driving bench-
 260 marks [3, 1, 43], leaving highway, rural, adverse-weather, and night-time conditions underrepre-
 261 sented in our characterization of the OL-CL gap. Extending BridgeSim’s scenario coverage to more
 262 diverse logs is a natural follow-up, though we expect the structural argument about objective mis-
 263 match to persist since it follows from the OL training objective rather than from scene composition.

264 8 Conclusion

265 In this work, we analyze the OL-CL gap in E2E autonomous driving and show that OL optimization
 266 does not guarantee safe and robust behavior in long-horizon CL simulation. Our results indicate
 267 that scaling OL training alone is insufficient for realistic deployment; instead, targeted test-time
 268 adaptation must address both observation and policy modules to achieve robust CL behavior. We
 269 further argue that current OL benchmarks on E2E driving is saturated, where OL metrics fail to
 270 reliably predict real-world-driving performance. Future progress should therefore prioritize realistic
 271 CL evaluation and closed-loop objective-aligned policy learning.

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459 Appendix

460 A More Related Works

461 **E2E Driving Policy Adaptation.** Adaptation of end-to-end (E2E) driving policies has long been
462 studied under the broader problem of covariate shift in imitation learning. Behavior cloning suf-
463 fers from compounding errors when policies encounter off-distribution states, motivating interactive
464 learning methods such as DAgger that address behavioral distribution shift through on-policy data
465 aggregation [51, 52]. In driving-specific settings, prior studies confirm that large-scale OL imitation
466 learning often overfits to static expert trajectories and fails under long-horizon or reactive interac-
467 tions [53, 10]. Orthogonal to behavioral shift, sim-to-real transfer research highlights observational
468 distribution shift as a critical failure mode for E2E policies [22]. While domain randomization and
469 representation learning improve robustness, they do not explicitly guarantee planning-oriented fi-
470 delity across heterogeneous simulators. Recent OL benchmarks enable scalable evaluation but have
471 been shown to correlate weakly with CL reliability, underscoring the limitations of static evalua-
472 tion protocols [3, 2]. In contrast to prior work that addresses perception or policy shift in isolation,
473 our work provides in-depth decomposition of the OL-CL gap and studies their interaction under
474 controlled simulator-induced transition changes.

475 B More Details on BridgeSim Platform

476 B.1 Motivation

477 BridgeSim platform is a unified cross-simulator closed-loop simulation platform to enable E2E poli-
478 cies to be evaluated across simulators, as illustrated in Fig. 6. We argue that existing benchmarking
479 testbeds are insufficient for a rigorous and unified analysis of the OL-CL gap in E2E autonomous
480 driving. The limitations of current E2E driving benchmarks can be categorized into three primary
481 deficiencies: 1) open-loop evaluations in existing benchmarks [3, 2] often exhibit only a superficial
482 correlation with short-term closed-loop performance and fail to serve as reliable indicators of long-
483 horizon closed-loop capabilities; 2) current closed-loop benchmarks frequently suffer from a lack
484 of multi-view temporal consistency [26], inaccurate annotations for observationally consistent traf-
485 fic elements [27], or a lack of environmental diversity and agent behaviors necessary to stress-test
486 generalizability [6] (see Appendix E for more details); and 3) policies transferred between simula-
487 tors often encounter inconsistent vehicle dynamics and controller configurations, inducing artificial
488 performance drops that obscure the true OL-CL gap.

489 These systematic deficiencies necessitate the design of the BridgeSim platform, which is engineered
490 to be controllable for identifying modular changes and consistent across evaluation modes and met-
491 rics. To this end, BridgeSim leverages the MetaDrive [25, 54] simulation engine to resolve the
492 controllability and inconsistency issues prevalent in previous benchmarks. As illustrated in Fig. 6,
493 BridgeSim ensures high-fidelity closed-loop simulation through a modularized architecture imple-
494 menting: 1) a unified structural framework for policy transfer to maintain state-action compatibility
495 across domains; 2) unified scenario protocols that enable precise, repeatable environment replay
496 and controllable agent interactions; and 3) unified evaluation modes and metrics to ensure fair and
497 accurate comparisons.

498 B.2 Unified Structural Invariance in Policy Transfer

499 To quantify the OL-CL gap, it is imperative to decouple the policy’s decision-making logic from
500 confounding systemic variables. To establish an equitable diagnostic framework, BridgeSim en-
501 forces two levels of structural invariance:

502 *1) Observational Alignment:* We standardize sensor configurations, including camera extrinsics and
503 intrinsics, across all evaluated benchmarks. This standardization ensures that performance failures
504 are strictly attributable to the policy’s representational robustness rather than superficial geomet-

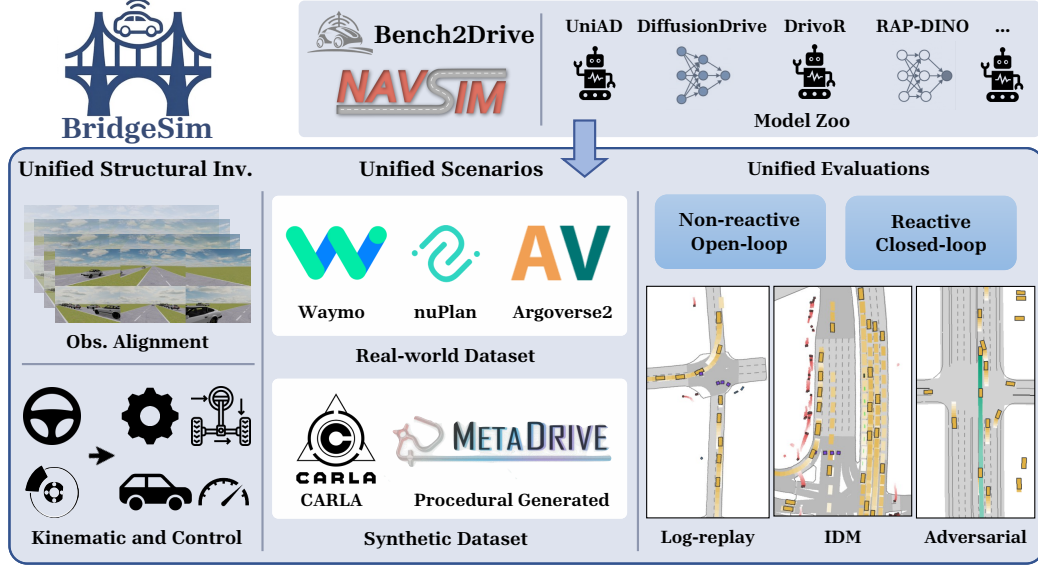


Figure 6: Illustration of BridgeSim Platform. BridgeSim supports unified closed-loop simulation of policies pretrained from any other simulators, enabling us to perform rigorous decomposition analysis.

505 ric mismatches in the input space Ω . Since the simulator maintains rigorous physical properties,
 506 we further enforce multi-view temporal consistency, addressing a common deficiency in existing
 507 benchmarks. Sensor observations o_t are synthesized via geometric projection, ensuring that o_t re-
 508 mains spatially and temporally consistent with the underlying state s_t across all camera frames and
 509 timesteps. This approach eliminates spurious distribution shifts, thereby ensuring that policy fail-
 510 ures are rooted in decision-making errors rather than simulation artifacts. Representative failure
 511 cases of existing generative and Gaussian Splatting-based simulation paradigms are detailed in the
 512 Appendix E.

513 2) *Kinematic and Control Alignment*: We enforce consistency across vehicle dynamics and control
 514 execution layers to isolate the sources of performance degradation. By standardizing the transition
 515 dynamics $\mathcal{P}$ during the execution of predicted waypoints, we ensure that the measured OL-CL gap
 516 reflects fundamental planning-level failures rather than downstream actuation discrepancies. To en-
 517 sure our evaluation reflects standard practice, we adopt controllers that are ubiquitous across diverse
 518 real-world autonomous driving platforms [55]; specifically, we utilize a standard PID formulation
 519 for longitudinal control and a Pure Pursuit [56] controller for lateral tracking. Furthermore, Brid-
 520 geSim maintains strict structural invariance by explicitly decoupling these vehicle dynamics from
 521 the rendering pipeline.

522 B.3 Unified Scenario Protocols

523 **Scenario Descriptions.** We define a unified, serialized data schema designed to encapsu-
 524 late all static and dynamic elements necessary to reconstruct any driving episode. To ensure
 525 cross-platform compatibility, the converters map diverse source data from real-world logs (e.g.,
 526 Waymo [43], nuScenes [1]) to synthetic environments (e.g., CARLA [6]) into a standardized
 527 ScenarioDescription format.

528 This centralized data structure contains three primary components:

529 1) *Map features*. BridgeSim ingests diverse real-world datasets, including Waymo [43], nuPlan [12],
 530 nuScenes [1], via a standardized protocol to extract map geometry. This geometry is discretized and
 531 stored as dense point sequences, each assigned specific type attributes. We subsequently vectorize
 532 HD map elements, such as lanes, boundaries, crosswalks, and stop lines, into a unified polyline

533 representation. By ensuring the precise spatial alignment of these layouts within the simulation en-
534 vironment, we eliminate rendering artifacts that might otherwise distort lane properties, maintaining
535 higher fidelity and consistency than existing benchmarks.

536 2) *Object tracks*. Dynamic agents are stored as temporal tracks containing state sequences, includ-
537 ing *position*, *heading*, *velocity*, and *dimensions* alongside validity masks. To address the varying
538 sampling rates across different datasets, we implement an interpolation pipeline that up-samples tra-
539 jectories. This process utilizes linear interpolation for position and velocity, while applying angular
540 interpolation for heading to ensure smooth kinematic playback during simulation.

541 3) *Dynamic map states*. Traffic light states are decoupled from the static map geometry. We map
542 individual traffic signals to specific lane IDs and store their state sequences (e.g., *STOP*, *WAIT*, *GO*).
543 This mapping enables the simulator to dynamically update the lane signaling during replay, ensuring
544 that the behavioral constraints of the agents remain in sync with the visual environment.

545 **Traffic Modes.** BridgeSim provides an API for the dynamic configuration of agent behaviors to
546 stress-test policies under varying levels of interactivity. We allow for the following agent behavior
547 modes:

548 1) *Log-replay mode*. In this mode, background agents strictly adhere to their recorded trajectories
549 from the source dataset. The simulator essentially replays the preset vehicle tracks. While this pre-
550 serves the exact real-world context, these agents remain non-reactive to the ego-vehicle’s deviations.

551 2) *IDM mode* [44]. To bridge the gap between static datasets and closed-loop interaction, we re-
552 place log-replay policies with an Intelligent Driver Model (IDM). While background vehicles are
553 initialized at their ground-truth poses, their subsequent behaviors are governed by heuristic policies.
554 These agents adhere to lane geometry, maintain safety gaps, and actively respond to the ego-vehicle’s
555 presence.

556 3) *Adversarial mode*. We provide an interface for injecting adversarial agents [45] into log-replay
557 scenarios. This enables the generation of safety-critical edge cases, challenging ego-policies to react
558 instantaneously to avoid imminent collisions.

559 **Coordinate Systems.** A critical challenge in evaluating policies across different simulators is the
560 misalignment of coordinate systems (e.g., global vs. local, left-handed vs. right-handed). BridgeSim
561 enforces a unified right-handed coordinate system (RHS) centered on a local scenario anchor. We
562 employ the following techniques to ensure coordinate system alignment:

563 1) *Chirality normalization*. For datasets using left-handed systems (e.g., CARLA, Bench2Drive),
564 we apply a basis transformation during conversion. Orientation angles (yaw, roll) are inverted ac-
565 cordingly to maintain kinematic consistency.

566 2) *Spatial anchoring*. To mitigate floating-point errors inherent in large-scale global coordinates
567 (common in nuScenes and nuPlan), we normalize all spatial data relative to a scenario anchor, de-
568 fined as the ego-vehicle’s position at $t = 0$. All map features and trajectories are translated so that
569 the map origin aligns with this anchor, ensuring numerical stability for the planner and simulator
570 physics engine.

571 **Time Lengths.** BridgeSim decouples the simulation horizon from the source simulators by sup-
572 porting long-term simulation durations determined by the full extent of the source log. The scenario
573 duration can also be manually configured to a variable length comprising a sub-duration of the full
574 source scenario. To accommodate the heterogeneous sampling rates of these diverse datasets, we
575 implement a standardized trajectory interpolation protocol. This pipeline resamples source signals
576 to the simulator’s native 10Hz control frequency using linear and angular interpolation. This ensures
577 consistent kinematic evaluation across varying time steps without artificially truncating the scenario.

578 B.4 Unified Evaluation Modes and Metrics

579 **Evaluation modes.** To rigorously assess the capability of driving agents to generalize from offline
580 datasets to online deployment, BridgeSim implements a unified evaluation pipeline. This pipeline

581 supports both *non-reactive open-loop evaluation* and *reactive closed-loop evaluation*, within a
 582 shared environment $\mathcal{E}$. Let $\mathcal{S}$ denote the state space and $\mathcal{A}$ the action space. At time t , the agent
 583 observes state $s_t \in \mathcal{S}$ and generates a planned trajectory $\tau_t = \pi(s_t)$.

584 *1) Non-reactive open-loop simulation.* In open-loop mode, the simulation decouples the ego-
 585 agent’s trajectory planning from the environment’s state transitions. The environment dynamics are
 586 strictly governed by the ground-truth trajectories from the source dataset. In this mode, the ego-
 587 vehicle’s state transition is forced to match the logged trajectory $\mathcal{T}_{log}$ independent of the policy
 588 output. Specifically, the state transition is defined as $s_{t+1} = f(s_t, a_t^*)$, where a_t^* denotes the expert
 589 action recorded in the log. The model predicts a trajectory $\hat{\tau}_t$ at every timestep, which is trans-
 590 formed from the ego-frame to the global simulation frame for evaluation against the static map and
 591 dynamic agents (replayed from logs). This mode isolates the planner’s reasoning ability without the
 592 confounding factor of cumulative control error.

593 *2) Reactive closed-loop simulation.* In closed-loop mode, the evaluation focuses on the agent’s
 594 ability to robustly control the vehicle over long horizons. The simulation environment is config-
 595 urable to support either a *fully reactive* paradigm, where background traffic behavior is governed
 596 by Intelligent Driver Models (IDM) that dynamically respond to the ego-agent’s movements, or a
 597 *semi-reactive* paradigm, where background traffic strictly adheres to the original logged trajectories
 598 while the ego-vehicle remains physically controlled by the policy.

599 **Evaluation metrics.** To quantify driving performance within the BridgeSim environment, we im-
 600 plement a set of open-loop and closed-loop metrics following previous protocols [3, 2, 27, 26, 57].
 601 For non-reactive open-loop simulations, we adopt the EPDMS framework [3, 2, 57]. For reac-
 602 tive closed-loop simulations, we incorporate a composite metric inspired by the HUGSIM [27] and
 603 DriveArena [26] Driving Score.

604 *1) Open-loop Metric: Extended Predictive Driver Model Score (EPDMS).* The **EPDMS** score is
 605 implemented as a weighted combination of safety-critical constraints and driving quality features, as
 606 shown in Table 3. Unlike L2 displacement errors, **EPDMS** penalizes physically infeasible or unsafe
 607 plans even if they mimic the log. At every frame, we transform the model’s predicted trajectory
 608 into the simulation frame and score it against the static and dynamic map features. The final score
 609 $S_{EPDMS} \in [0, 1]$ for a given frame is defined as the product of critical boolean constraints $\mathcal{C}$ and a
 610 weighted sum of soft metrics $\mathcal{F}$:

$$S_{EPDMS} = \left(\prod_{c \in \mathcal{C}} \mathbb{I}_c \right) \cdot \left(\frac{\sum_{f \in \mathcal{F}} w_f \cdot v_f}{\sum_{f \in \mathcal{F}} w_f} \right), \quad (8)$$

611 where $\mathbb{I}_c \in \{0, 1\}$ is the indicator function for constraint compliance, $v_f \in [0, 1]$ is the normalized
 612 feature score, and w_f is the corresponding weight.

613 *2) Closed-loop Metric: BridgeSim Driving Score.* For reactive closed-loop simulation, we define
 614 the final driving score S_{CLS} as the product of the agent’s global Route Completion (**RC**) and the
 615 temporal average of the frame-level scores ($S_{EPDMS}^{(t)}$):

$$S_{CLS} = R_c \times \frac{1}{T} \sum_{t=1}^T S_{EPDMS}^{(t)}, \quad (9)$$

616 where $R_c \in [0, 1]$ represents the percentage of the route completed by the agent relative to the expert
 617 driver’s path or the goal destination, T is the total number of frames in the simulation episode, and
 618 $S_{EPDMS}^{(t)}$ is the EPDM score at time step t . Note that the frame-level ego progress score (**EP**)
 619 from open-loop EPDM scoring is removed, as progress is now explicitly measured by the distance
 620 traveled to the goal in the simulator [27]. Crucially, for the closed-loop setting, the set of soft
 621 features $\mathcal{F}$ is modified to exclude displacement error or progress-based terms, focusing solely on
 622 driving quality and safety constraints. This formulation ensures that an agent achieves a high score
 623 only if it completes the route while consistently maintaining safe and comfortable driving behavior
 624 throughout the episode.

Table 3: Descriptions of subscores of EPDMS.

Metric	Description
Critical Constraints ($\mathcal{C}$)	
No At-Fault Collisions (NC)	Ego polygon $\mathcal{P}_{ego}$ must not intersect any replayed object polygons $\mathcal{P}_{obj}$.
Drivable Area Compliance (DAC)	Entire horizon must satisfy $\mathcal{P}_{ego} \subset \mathcal{M}_{drivable}$ within the HD map.
Traffic Light Compliance (TLC)	Agent cannot cross stop lines associated with red light states.
Driving Direction Compliance (DDC)	Heading deviation $ \theta_{ego} - \theta_{lane} $ must remain within $\pm\pi/2$.
Quality Features ($\mathcal{F}$)	
Ego Progress (EP)	Represents the agent progress as a ratio to an approximated safe upper bound.
Lane Keeping (LK)	Checks the ego-vehicle follows the centerline of its current lane and avoids lingering between adjacent lanes.
Time-to-Collision (TTC)	Penalizes trajectory $\hat{\tau}$ if the projected time-to-collision is too low.
History Comfort (HC)	Evaluates the planner’s predicted trajectory with a short segment of historical motion from the human driver.
Extended Comfort (EC)	Limits kinematic derivatives for passenger comfort: acceleration $a \leq 4.89$, jerk $j \leq 8.37$, and yaw rate $\dot{\theta} \leq 0.95$.

625 B.5 Sensor Models

626 BridgeSim inherits the sensor simulation capabilities of the MetaDrive engine [25, 54], which pro-
 627 vides a configurable suite of ego-centric sensors rendered through the underlying Panda3D graph-
 628 ics pipeline. The platform supports multi-view RGB cameras with arbitrary extrinsic placement
 629 and configurable intrinsic parameters (resolution, field-of-view, and focal length). Beyond RGB,
 630 MetaDrive natively exposes depth and semantic segmentation buffers, instance segmentation masks,
 631 top-down BEV rasterizations, and 3D LiDAR.

632 B.6 Sim-to-Sim Evaluation Results across Simulators

633 B.6.1 Supported Model Zoo

634 Sim-to-sim transfer refers to the training and testing environments are different simulation environ-
 635 ments in order to examine policy’s transferability in novel simulation environments: for example,
 636 NAVSIM [3, 2] is a simulation environment with real-world observations, whereas Bench2Drive [6]
 637 is a simulation environment with game-engine based observations. We include a diverse set of pub-
 638 licly available end-to-end driving models for zero-shot cross-simulator evaluation. Unless otherwise
 639 specified, we use the official checkpoints released by the original authors or benchmark maintain-
 640 ers. Note that new models could be easily incorporated into BridgeSim following the instructions in
 641 Appendix B.7.

642 **UniAD** [35] is a unified end-to-end autonomous driving framework that jointly models percep-
 643 tion, prediction, and planning through a query-based transformer architecture. We use its official
 644 Bench2Drive checkpoint in our evaluation.

645 **Vectorised Autonomous Driving (VAD)** [36] is a vectorized end-to-end driving method that repre-
 646 sents map elements and traffic agents with structured queries rather than dense raster features. We
 647 use its official Bench2Drive checkpoint.

648 **TransFuser** [37] is a multi-modal end-to-end driving model that fuses RGB camera and LiDAR
 649 observations for waypoint prediction. We use the released NAVSIM-trained checkpoint in our eval-
 650 uation.

651 **DiffusionDrive** [38] formulates autonomous driving as a generative planning problem using a trun-
 652 cated diffusion process with anchor trajectories. The released checkpoint used in our evaluation is
 653 trained on NAVSIM.

654 **DiffusionDriveV2** [39] extends DiffusionDrive by incorporating reinforcement learning into the
655 trajectory denoising process. We use the released NAVSIM-trained checkpoint.

656 **LEAD** [42] studies learner-expert asymmetry in imitation learning and builds on a TransFuser-v6
657 driving architecture. The released checkpoints include CARLA-based models for Leaderboard 2.0 /
658 Bench2Drive-style evaluation as well as a NAVSIM checkpoint under the LTFv6 variant.

659 **DrivoR** [5] is a transformer-based end-to-end planner built on a pretrained DINOv2 visual encoder
660 with token compression and trajectory scoring modules. The released model used in our evaluation
661 is trained on NAVSIM.

662 **Rasterization Augmented Planning (RAP)** [4] is a rasterisation-based augmentation framework
663 for end-to-end driving that improves training with counterfactual rasterised views and raster-to-real
664 feature alignment. The official checkpoint used in our comparison is trained on NAVSIM.

Table 4: Quantitative comparison of E2E driving policies evaluated on BridgeSim-NavHard. All results are reported with a replan rate of 5 and a simulation horizon of 8 seconds. We use $rr = 5$ to align with the temporal discretization commonly used for training on NAVSIM/OpenScene, where the dataset is provided at 2 Hz (i.e., $\Delta t = 0.5$ s).

Model	Pretrained Sim	DS	EPDMS	RC	NC	DAC	DDC	TLC	TTC	LK	HC	EC
Human Expert	N/A	91.14	91.96	98.86	98.18	99.23	99.76	98.62	94.55	90.22	99.10	71.44
UniAD (CVPR'23)	Bench2Drive	31.00	57.55	55.25	93.15	71.75	99.76	99.61	81.13	97.70	74.25	30.04
VAD (ICCV'23)	Bench2Drive	26.25	57.60	46.76	93.29	71.22	99.76	99.68	82.31	98.67	71.13	27.67
DiffusionDrive (CVPR'25)	NAVSIM	43.29	69.70	61.30	95.38	81.48	99.76	98.81	87.47	91.57	91.64	39.02
DiffusionDriveV2 (arXiv)	NAVSIM	29.81	62.01	45.85	93.48	79.75	99.76	98.83	86.93	84.11	71.38	20.62
RAP (ICLR'26)	NAVSIM	52.08	72.41	70.74	95.63	85.24	99.76	99.18	88.14	84.69	92.84	48.91
LEAD (CVPR'26)	NAVSIM	55.61	71.49	76.93	95.62	90.73	100.00	98.95	88.22	81.61	94.88	45.13
TransFuser (PAMI'23)	NAVSIM	41.46	68.17	61.28	95.73	78.99	99.76	98.63	88.79	89.25	95.06	49.25
LTF (NeurIPS'24)	NAVSIM	50.10	69.29	70.64	95.49	78.80	99.53	98.51	88.61	90.28	95.02	58.27
DrivoR (CVPR'26)	NAVSIM	42.79	66.54	64.66	95.41	76.34	100.00	99.46	95.41	94.37	96.99	66.69
TTA (DiffusionDrive)	NAVSIM	67.84	79.77	82.54	96.48	87.03	100.00	99.01	96.07	91.85	36.18	33.72
TTA (DiffusionDriveV2)	NAVSIM	61.46	76.86	77.86	95.83	86.80	100.00	99.17	92.93	91.51	26.22	24.32
TTA (RAP)	NAVSIM	80.92	91.68	87.06	96.95	90.14	100.00	98.90	98.07	94.72	46.18	54.60

665 B.6.2 Sim-to-Sim Evaluation Results

666 Our evaluation in Table 4 reveals that most open-source models suffer from the OL-to-CL deploy-
667 ment gap when integrated into the BridgeSim platform. We observe that models pretrained on
668 synthetic simulation domains frequently encounter simulation overfitting across both visual and be-
669 havioral aspects. Specifically, models trained on Bench2Drive exhibit a substantial performance
670 degradation. This disparity underscores a significant sim-to-sim gap between CARLA’s synthetic
671 rendering and BridgeSim’s real-world replayed scenarios. On the other hand, NAVSIM-pretrained
672 models demonstrate better performance, as their training environment relies on real-world traffic
673 behaviors. These findings emphasize that, besides visual fidelity, behavioral realism is also critical
674 when training autonomous agents for robust simulation performance. The results show that strong
675 in-domain performance doesn’t always translate to broader generalizability.

676 B.7 Extensible Design for Incorporating New Maps, Models, and Traffic Modes

677 BridgeSim possesses high backward-compatibility and extensibility to incorporate new maps, mod-
678 els, and traffic modes, encouraging open-source contributors to independently augment the bench-
679 mark with novel driving environments, state-of-the-art planning models, and alternative traffic be-
680 haviors without modifying any shared evaluation logic.

681 **Incorporating New Maps.** To integrate a new map, contributors only need to implement the
682 `converter` module and integrate the new maps into the scenario description protocol. This module
683 provides dedicated, dataset-specific pipelines for each supported source. Each pipeline implements
684 a conversion function that transforms raw, dataset-native records into a standardized scenario de-
685 scription dictionary. By invoking the shared serialization utility within this function, the resulting

scenario directory becomes automatically discoverable by the environment manager during evaluation, requiring no modifications to downstream evaluation logic.

Incorporating New Models. Contributor could use the adapter module to incorporate new planners for evaluations. We provide an interface through the abstract base class `BaseModelAdapter`, which has four core functions to enable simulation on BridgeSim: `load_model()` for checkpoint initialization and device mapping; `prepare_input()` for transforming raw multi-camera images and ego-state vectors into the model’s native input format; `run_inference()` for executing the forward pass; and `parse_output()` for extracting a predicted waypoint trajectory of shape $N \times 2$ in the ego-vehicle coordinate frame. Once the model adapter is implemented, it is registered by adding a single entry to the `create_model_adapter()` factory, after which the new model is immediately ready for evaluations across all loaded maps and traffic modes.

Incorporating New Traffic Modes. Traffic behavior is governed by the `EnvironmentManager` component, which maps symbolic traffic-mode identifiers to their corresponding MetaDrive simulation configurations. To introduce a new traffic mode, such as a learning-based traffic generation model or an adversarial agent policy, contributors simply define a new mode identifier, extend the internal configuration resolver to emit the necessary MetaDrive flags, and optionally provide a custom agent policy class.

C Implementation Details on TTA Framework

C.1 Implementation Details on Observational Calibrator

The observational calibrator is designed to align latent representation distributions $\mathcal{D}_{\text{source}}$ and $\mathcal{D}_{\text{target}}$ via a flow-matching framework. To construct the necessary feature pairs for training, we utilize the NAVSIM dataset [3, 2], replaying and rendering all annotated frames to generate paired image observations between the source and target domains, as shown in Fig. 7. To process these input observations, we first employ a small encoder that transforms raw scene features into a regularized latent space $\mathcal{Z}$. We then parameterize the vector field $v_\theta(z, t)$ operating on this space using a Diffusion Transformer (DiT) [58] architecture. Specifically, we employ the *DiT-L* configuration with a patch size of 2 to process the latent features, ensuring high-resolution alignment across the domain shift.

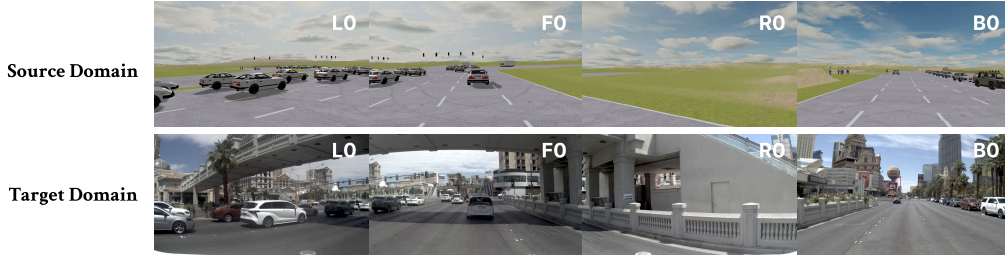


Figure 7: Visualization of observation pairs in source and target domains.

Following [48], we optimize the calibrator using a multi-objective loss function. In addition to the standard flow-matching objective, the intermediate latent space $\mathcal{Z}$ is explicitly regularized via an **encoder loss** and a **KL divergence loss** to preserve semantic consistency and maintain a well-behaved manifold. The total training objective is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{FM}} + \lambda_{\text{enc}} \mathcal{L}_{\text{enc}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}}, \quad (10)$$

where each component is defined as follows:

- **Flow-Matching Loss ($\mathcal{L}_{\text{FM}}$):** We adopt a linear probability path to learn the transport map between source and target distributions. The loss is defined as:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, q(z_t | z_0, z_1)} [\|v_\theta(z_t, t) - (z_1 - z_0)\|^2], \quad (11)$$

where $z_0 \sim \mathcal{D}_{\text{source}}$ and $z_1 \sim \mathcal{D}_{\text{target}}$.

- **Encoder Loss ($\mathcal{L}_{\text{enc}}$):** We employ a categorical contrastive loss between source and target feature pairs to regularize the small encoder. This term ensures that latent features corresponding to similar underlying states are clustered together while pushing dissimilar states apart. This prevents mode collapse during the transport process and preserves the discriminative structure of the latent space necessary for the downstream policy π_ϕ .
- **KL Loss ($\mathcal{L}_{\text{KL}}$):** To maintain a structured and smooth latent manifold, we apply a Kullback-Leibler divergence penalty against a standard normal prior $\mathcal{P}(z) = \mathcal{N}(0, \mathbf{I})$:

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}(q(z|o) \parallel \mathcal{P}(z)). \quad (12)$$

This variational regularization ensures that the flow-matching trajectory remains within a well-behaved region of the latent space $\mathcal{Z}$.

During training, we use the AdamW [59] optimizer with a learning rate of 1×10^{-4} and weight decay of 0.01. At test time, the calibrated representations are obtained by integrating the learned ODE from $t = 0$ to $t = 1$ using a first-order Euler solver.

C.2 More Implementation Details on Test-time Policy Adaptation

To operationalize the truncated action-value estimator defined in Section 5.2.1, we employ Monte Carlo sampling to generate and evaluate future candidate rollouts over the horizon H . The empirical reward function $R_k(s_t, \mathbf{a}_t)$ evaluates each candidate prefix by computing **EPDMS**. Assuming the subscore definitions established in Appendix B.4, we detail the computational mechanisms enabling their exact and estimated evaluation during real-time closed-loop execution.

The prefix reward is mathematically formulated as a gated sum of the continuous Quality Features ($\mathcal{F}$), strictly conditioned on the satisfaction of the Critical Constraints ($\mathcal{C}$):

$$R_k(s_t, \mathbf{a}_t) = \mathbb{I}_{\mathcal{C}} \cdot \sum_{f \in \mathcal{F}} w_f f(s_{t:t+k}, \mathbf{a}_{t:t+k}) \quad (13)$$

where $\mathbb{I}_{\mathcal{C}} \in \{0, 1\}$ acts as a hard safety filter. If a candidate trajectory violates any critical constraint, $\mathbb{I}_{\mathcal{C}} = 0$, immediately zeroing out the trajectory’s expected value.

Kinematic Comfort. Because the test-time adaptation mechanism operates directly on spatial-temporal plans, we have explicit access to both the historical states and the predicted future waypoints of the ego vehicle. Consequently, we do not need to rely on transition approximations; the exact comfort score is computed deterministically by extracting the high-frequency kinematic derivatives (acceleration, jerk, and yaw rate) directly from the sequence of waypoints.

Semantic and Map Awareness. Evaluating map-dependent metrics, such as Drivable Area Compliance (**DAC**), Driving Direction Compliance (**DDC**), Traffic Light Compliance (**TLC**), and Lane Keeping (**LK**), requires grounding the generated waypoints in the environment’s topology. To achieve this, we assume access to an accurate localization model and a robust mapping model, such as High-Definition (HD) Map or other online map generative models. The candidate ego-polygon $\mathcal{P}_{\text{ego}}$ is projected at each timestep and geometrically queried against the generated real-time semantic layouts to compute both constraint satisfaction and routing efficiency.

Dynamic Safety Estimation. Evaluating interaction safety, specifically No At-Fault Collisions (**NC**) and Time-to-Collision (**TTC**), requires predicting the stochastic futures of other actors. We assume access to the current estimated velocity and acceleration profiles of surrounding agents. Using these profiles, we kinematically propagate their bounding boxes $\mathcal{P}_{\text{obj}}$ over the planning horizon to check for intersections with the ego trajectory and to compute the time-to-collision margins.

Two-Dimensional Surrogate Safety Measures. To rigorously evaluate the spatial-temporal proximity to potential collisions across complex driving environments, we utilize a two-dimensional Surrogate Safety Measure (SSM) framework. The foundational metric in this domain is the Time-to-Collision (TTC), which evaluates the temporal proximity to a conflict based on the projected Distance-to-Collision (*DTC*). Operating under a classical assumption of constant velocity at the

instantaneous moment of evaluation [60], the base 2D-TTC is defined using the relative velocity vector $\mathbf{v}_{ij}$ between the host vehicle i and an interacting agent j :

$$TTC = \frac{DTC}{\|\mathbf{v}_{ij}\|} \quad (14)$$

While computationally straightforward, the standard constant-velocity assumption exhibits significant drawbacks in dynamic, real-world driving scenarios. Specifically, it fails to capture slow, conscious interactive behaviors, such as yielding, braking, or evasive maneuvering, often leading to inaccurate risk assessments during higher-order kinematic interactions [61, 62].

To address these limitations, we adopt the Modified Time-To-Collision (MTTC) as our primary safety evaluation metric. The MTTC relaxes the constant-velocity constraint by incorporating the relative acceleration vector, $(\mathbf{a}_i - \mathbf{a}_j)$, thereby yielding a higher-fidelity, second-order temporal projection. By solving the kinematic displacement equation, the MTTC is formulated as:

$$MTTC = \frac{-\|\mathbf{v}_{ij}\| \pm \sqrt{\|\mathbf{v}_{ij}\|^2 + 2(\mathbf{a}_i - \mathbf{a}_j)DTC}}{\mathbf{a}_i - \mathbf{a}_j} \quad (15)$$

Consistent with established collision warning logic, the final MTTC is determined by extracting the minimum positive real root from this quadratic formulation. Furthermore, by structuring the vehicle states into matrices, we enable the parallel computation of the MTTC across all i, j pairs simultaneously. The resulting indicator sets are subsequently filtered according to the behavioral thresholds established by Ozbay et al. [63], ensuring both high-fidelity risk assessment and computational efficiency for large-scale, multi-agent evaluation.

D More Experimental Details

D.1 Closed-loop Evaluation Setting

Scenario Descriptions. We convert three real-world datasets into BridgeSim platform. **BridgeSim-NavHard** is derived from the NAVSIM benchmark [3], whose logs originate from the nuPlan dataset [12] at 2 Hz. Agent trajectories are up-sampled to 10 Hz via linear interpolation, and HD maps are extracted from the nuPlan map API with a large spatial buffer to cover the full ego trajectory. The converted scenes contain 421 scenarios. **BridgeSim-WOMD** is converted from the Waymo Open Motion Dataset [64], which provides 9-second scenarios at 10 Hz in a world-frame coordinate system. All agent and map positions are re-centered at the ego vehicle’s initial position, and ego velocity is rotated into the vehicle’s local frame. The converted scenes contain 400 scenarios. **BridgeSim-nuScenes** is built on the nuScenes dataset [1], which runs at 2 Hz. The converted scenes contain 143 scenarios. Camera images are re-rendered in MetaDrive at 10 Hz, sampled at every fifth step to match the original keyframe rate.

Simulation Settings. All models are evaluated under simulation operating at a default timestep (`sim_dt=0.1s`). Inference is executed at a specified `replan_rate`, with cached waypoints consumed between replan intervals. An initial replay phase is performed before evaluation begins to provide historical temporal buffer for agents. For multi-proposal models, a trajectory scorer selects the final plan within a set of trajectory candidates. For running TTA module, an observational calibration module is applied to reduce the domain gap between simulation and real-world training data via a flow matching network with a fixed number of sampling steps.

D.2 Details on Empirical Study in Section 4.2

In Fig. 2 from Section 4.2, the Average Displacement Error (ADE) comparisons are obtained by evaluating the policy’s predicted trajectories over a 4-second horizon against the ground-truth expert demonstrations. To quantify the prediction accuracy accounting for multi-modal outputs, we compute the Minimum Average Displacement Error (minADE) by evaluating the closest predicted trajectory candidate to the ground truth. Let T denote the prediction horizon, K the number of

808 predicted trajectory candidates, $\mathbf{p}_t$ the ground truth spatial coordinates at time step t , and $\hat{\mathbf{p}}_t^{(k)}$ the
 809 predicted coordinates for the k -th candidate. The metric is formulated as:

$$\text{minADE} = \min_{k \in \{1, \dots, K\}} \frac{1}{T} \sum_{t=1}^T \left\| \hat{\mathbf{p}}_t^{(k)} - \mathbf{p}_t \right\|_2, \quad (16)$$

810 where $\|\cdot\|_2$ denotes the Euclidean (L_2) norm. Furthermore, the closed-loop performance metrics
 811 are evaluated using the *BridgeSim Driving Score* across the BridgeSim-NavHard scenarios.

812 D.3 Details on Empirical Study in Section 4.3

813 In the analysis presented in Fig. 3, we employ the *BridgeSim Driving Score* as the empirical real-
 814 ization of the return objective $J(\pi)$. To ensure a rigorous assessment, we evaluate a representative
 815 family of state-of-the-art E2E policies [37, 38, 39, 4, 5], reporting the upper and lower performance
 816 bounds to characterize the variance across different architectural paradigms. The experimental pro-
 817 tocols are defined as follows:

- 818 • Fig. 3(a): we evaluate policies over a fixed simulation horizon of $T = 80$ in BridgeSim-
 819 NavHard. To isolate the impact of the execution horizon k , we compare a baseline of fixed
 820 k -step prefixes against an “optimal execution” baseline derived via exhaustive search over
 821 the policy’s predicted trajectory set.
- 822 • Fig. 3(b): we contrast the performance of policies operating under two distinct selection
 823 criteria in BridgeSim-Navhard: 1) the **OL Objective**, which relies on the agent’s learned
 824 Q-value estimator, and 2) the **CL Objective**, where actions are selected via the ground-
 825 truth EPDMS metric to simulate an unbiased value function. The evaluation is done across
 826 ($T \in \{40, 80, 120, 160\}$).
- 827 • Fig. 3(c): we compute the Pearson correlation between open-loop and closed-loop metrics
 828 across the policies. The OL metric is obtained through non-reactive simulation ($T = 40$)
 829 following the NAVSIMV2 protocol [2] in BridgeSim-NavHard. In contrast, the CL metric
 830 is derived from the BridgeSim Driving Score across expanding simulation horizons ($T \in$
 831 $\{40, 80, 120, 160\}$) to observe the decay of predictive validity over time in BridgeSim-
 832 NavHard.

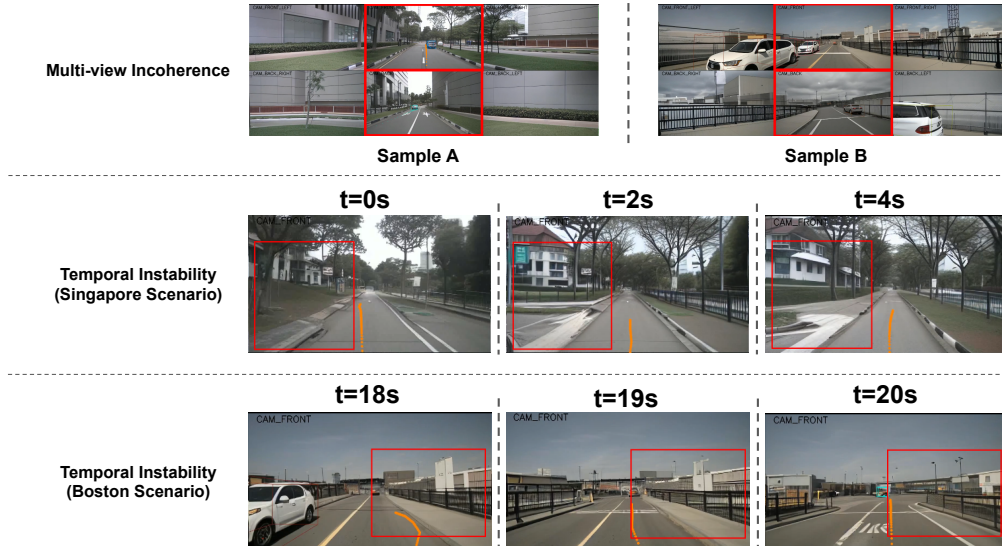


Figure 8: Qualitative analysis of simulation artifacts in DriveArena: multi-view incoherence (left) and temporal instability (right). Red boxes highlight inconsistencies that lead to policy failures or metric distortion.



Figure 9: Qualitative analysis of reconstruction artifacts and ghosting effects in HUGSIM. Red boxes highlight inconsistencies.

Table 5: Performance comparisons with different reactive traffic modes with complete sub-metric scores.

Model	IDM											Safety-Critical										
	DS	EPDMS	RC	NC	DAC	DDC	TL	TTC	LK	HC	EC	DS	EPDM	RC	NC	DAC	DDC	TL	TTC	LK	HC	EC
DiffusionDrive	45.30	72.95	59.93	95.41	85.47	100.00	98.36	89.73	89.83	92.31	40.82	38.68	68.16	54.77	93.62	83.86	100.00	98.36	84.97	90.71	89.56	39.13
DiffusionDriveV2	31.94	65.43	45.85	94.57	82.09	100.00	98.23	89.10	85.42	74.52	22.99	28.17	61.45	43.50	93.15	80.78	100.00	98.30	85.72	85.81	70.99	21.68
RAP	60.02	76.85	77.20	97.68	88.84	100.00	98.49	93.07	78.16	94.49	53.84	51.83	70.88	71.22	95.03	88.16	100.00	98.58	86.37	78.26	90.22	50.53
TTA (DiffusionDrive)	62.68 ± 17.4	84.36	72.28	96.85	95.63	100.00	98.98	91.57	92.43	87.95	40.13	55.02 ± 16.3	78.95	67.92	94.73	95.21	100.00	98.76	86.64	91.30	83.65	37.11
TTA (DiffusionDriveV2)	53.96 ± 22.0	77.47	67.89	96.08	90.53	100.00	99.43	89.46	89.98	85.46	29.90	47.64 ± 19.5	71.66	65.91	93.54	89.16	100.00	99.07	84.23	90.11	81.73	27.49
TTA (RAP)	79.07 ± 19.1	90.92	86.21	98.28	98.62	100.00	99.02	95.31	91.52	92.60	51.38	65.94 ± 14.1	82.01	78.66	95.30	96.19	100.00	99.14	87.71	90.26	88.18	48.07

D.4 Details on Experimental Results in Reactive Evaluation

We show the complete sub-metric scores in reactive evaluations results (see Table 2) in Table 5.

D.5 More Discussions of OL Training and Evaluations

Will RL finetuning on OL proxy reward benefit CL performance? Recent papers [39, 65] advocate for RL finetuning to enhance planning performance, primarily utilizing open-loop proxy rewards as the training objective. However, as shown in Table 2, our comparison between DiffusionDrive and DiffusionDriveV2 reveals a significant discrepancy between open-loop and closed-loop outcomes. Specifically, DiffusionDriveV2 exhibits a pronounced performance degradation in CL simulations while it obtains higher OL metrics in NAVSIM. This suggests that optimizing the OL objective doesn’t guarantee the optimization of the CL objective.

To what extent can we rely on OL metrics to predict CL performance? Fig. 3(c) indicates that while OL metrics maintain predictive validity for short horizons ($T = H$), this correlation decays rapidly as T exceeds H . This decaying trend indicates that while high OL scores signify that a policy possesses sufficient local planning proficiency and safety for in-distribution scenarios, they do not serve as a reliable predictor of CL performance in reactive or out-of-distribution environments. These findings suggest two critical takeaways: 1) scaling open-loop training data doesn’t statistically guarantee improvements for closed-loop performance, and 2) effective policies must prioritize objectives that approximate gold Q-value estimations to ensure long-horizon stability.

E Limitations of Existing E2E Closed-loop Benchmarks

Existing end-to-end closed-loop benchmarks, such as DriveArena [26] and HUGSIM [27], exhibit limitations stemming from *temporal and semantic inconsistencies* during simulation.

Generative Artifacts in DriveArena. As illustrated in Fig. 8, DriveArena displays notable simulation discrepancies. 1) *Multi-view Incoherence*: in the multi-view comparison, discrepancies in the color, length, and placement of lane markings between front and rear camera views (red boxes) can lead to oscillatory policy behavior due to contradictory observations. 2) *Temporal Instability*: as shown in the temporal sequences, road features such as intersections are inconsistently rendered or spontaneously modified across sequential timestamps (e.g., $t = 2s$ in Singapore and $t = 19s$ in Boston, marked by red boxes), resulting in unstable scene geometry.

Semantic Mismatch in HUGSIM. As shown in Fig. 9, 3D Gaussian Splatting-based methods like HUGSIM encounter challenges with dynamic actor integration. The Sample A & B highlight simulation artifacts and blurring around moving vehicles (red boxes), which are frequently misinterpreted as obstacles. Furthermore, the misalignment between visual rendering and the underlying HD map creates semantic inconsistency.

Structural Invariance in BridgeSim. In contrast, BridgeSim maintains strict structural invariance by decoupling vehicle dynamics from visual rendering. By leveraging the MetaDrive physics engine, the kinematic state s_t serves as the deterministic reference for the environment. Sensor observations o_t are generated via geometric projection rather than generative synthesis, ensuring spatial and temporal consistency across all camera frames and timestamps.

F Future Work

We expect that future research will focus on building more realistic evaluations and developing methods that enhance driving agents’ safety and robustness under real-world constraints:

1. E2E Closed-loop Benchmarks. Existing CL benchmarks remain limited by observational fidelity, temporal inconsistencies, and evaluation efficiency (see Appendix E). Future benchmarks should move toward unified, hybrid-engine frameworks that combine the controllability of game-engine simulation with the diversity of data-driven world models, enabling scalable yet realistic stress testing of long-horizon CL simulation.

2. E2E Closed-loop Training. Current OL pretraining methods lack error-recovery samples and feedback-driven learning objectives to teach a policy to safely navigate dynamic and reactive environments. Advancing CL training methods, such as scalable variants of RoAD [66], learning with world modeling, and reinforcement learning with reliable reward formulation, remains critical. In particular, defining and learning unbiased value functions that reflect true CL returns in real-world driving is a key open challenge.